Available online at www.sciencedirect.com

ScienceDirect

IFAC PapersOnLine 54-1 (2021) 49–54

Towards Using Digital Intelligent Assistants to Put Humans in the Loop of Predictive Maintenance Systems

Stefan Wellsandt*. Konstantin Klein* Karl Hribernik* Marco Lewandowski*
Alexandros Bousdekis Gregoris Mentzas** Klaus-Dieter Thoben*****

* BIBA - Bremer Institut für Produktion und Logistik GmbH at the University of Bremen, Hochschulring 20, 28359 Bremen, Germany (e-mail: {wel, kle, hri}@biba.uni-bremen.de)
** Information Management Unit (IMU), Institute of Communication and Computer Systems (ICCS), National Technical University of Athens (NTUA), 9 Iroon Polytechniou str., 157 80 Zografou, Athens, Greece (e-mail: {albous, gmentzas}@mail.ntua.gr)
*** University of Bremen, Faculty of Production Engineering, Badgasteinerstr. 1, 28359 Bremen, Germany (e-mail:tho @ biba.uni-bremen.de)

Abstract: Predictive maintenance systems are socio-technical systems where the interaction between maintenance personnel and the technical system is critical to achieving maintenance goals. Employees who use a predictive maintenance system should explore, modify, and verify their analysis and decision-making methods and rules. Conventional modes of interaction make this difficult since they are often hard to understand, obtrusive and unintuitive. Digital Intelligent Assistants (DIAs) provide fast, intuitive, and potentially hands-free access to systems through voice-based interaction and cognitive assistance. This paper introduces a novel approach to interact with predictive maintenance systems through DIAs. The aim is to integrate human knowledge more effectively into the predictive maintenance process to create a hybrid-intelligence system. In such systems, humans and computers complement and evolve together.

Keywords: Engineering Applications of Artificial Intelligence, Predictive Maintenance, Human-Automation Integration, Hybrid Intelligence Systems

1. INTRODUCTION

Reducing costs, improving production's reliability and efficiency while simultaneously taking sustainability and safety issues into account are pressing issues for all industries (Bousdekis *et al.*, 2018). Maintenance has a significant impact on all of these issues. Improvements in maintenance can lead to a 10-20% reduction of costs in related labour and materials, which together make up 15% of the total costs of a typical manufacturing company (McCarthy, Zinser and Spindelndreier, 2013). This potential means that maintenance needs to be considered an essential operation function for these organisations (Peng, Dong and Zuo, no date; Bousdekis *et al.*, 2015).

Maintenance strategies can be categorized as follows: 1) reactive maintenance, where maintenance is carried out after a fault occurs; 2) preventive maintenance, where maintenance is rigidly scheduled to prevent faults occurring; and 3) condition-based, predictive and prescriptive maintenance, where maintenance is planned to optimally take place based on data-driven diagnoses or predictions about an asset's condition (Bousdekis *et al.*, 2015). The increasing product complexity and customer demands for quality and reliability drive the trend towards predictive and prescriptive maintenance (Jardine, Lin and Banjevic, 2006). These approaches rely on sensors, programmable logic controllers (PLCs), Big Data, artificial intelligence (AI), and cloud computing. By analysing data generated by embedded sensors, assets can determine or predicted their condition and give appropriate maintenance recommendations. A predictive maintenance system's primary function is to identify an asset's current condition and accurately forecast its degradation. This information allows better planning of maintenance activities and the generation of optimal maintenance actions and schedules to avoid or mitigate asset or component failures.

Predictive maintenance systems are socio-technical hybrid-intelligence systems that rely on human knowledge to function optimally (Herrmann, 2020). Current concepts often lack an appropriate approach to include human maintenance actors in the decision-making process. While advanced visualisation technologies such as augmented reality (AR) are starting to be introduced, most operations still require complicated user interfaces on a computer or even require programming work. These barriers to involving humans need to be removed to realise the potential of predictive maintenance systems fully.

This paper proposes a novel approach to interact with predictive maintenance systems through **Digital Intelligent Assistants** (DIAs). The aim is to integrate human knowledge more effectively into the predictive maintenance process. The paper is structured as follows: Section 2 presents a predictive maintenance model to serve as a framework for identifying

10.1016/j.ifacol.2021.08.005

junctions at which human intervention in the process is advantageous. Section 3 introduces related work, which examines contemporary approaches to involving humans in predictive maintenance. Besides, it introduces the current state-of-the-art of DIAs. Sections 4 and 5 identify predictive maintenance functions that benefit from human intervention. It maps these findings to the predictive maintenance model, highlighting how DIAs can support these functions. Finally, it proposes a predictive maintenance system architecture that uses DIAs to realize the human-in-the-loop concept. Section 6 concludes this paper.

2. BACKGROUND AND MOTIVATION

Predictive maintenance systems should be understood as socio-technical systems (Herrmann, 2020) since the interaction between human maintenance actors and the technical system is critical to achieving maintenance goals. Analysis results and automated decisions should complement the involved people's knowledge and experience, supporting them to perform their work faster, accurately, and easier.

Encoding maintenance knowledge and experience into a predictive maintenance system is challenging. For example, not all of the knowledge required may be available, especially for new assets or production systems, since experience has yet to be made. In consequence, interpretations of diagnoses and prognoses may be inaccurate, resulting in inappropriate maintenance recommendations. Maintenance experts need to reconfigure the system continuously to integrate new experiences. Similarly, personnel may improve maintenance practices over time and with experience, resulting in new rules for the decision-making component.

Rather than repeatedly reconfiguring algorithms, rules, and parameters, it would be preferable to involve humans in the predictive maintenance process to communicate new knowledge to the system quickly. By creating a dialogue between the system and its users, the knowledge encoded in the system could continuously be improved, creating a self-learning loop that optimises the system and provides more benefit to maintenance actors.

The theory of **hybrid intelligence systems** suggests that human input to AI systems can be used to prevent mistakes and shortcomings and help them learn what decision is best to take for a given problem (Kamar, 2016). Humans and AI's complementary strengths can work together to co-evolve by learning from each other and thus improving over time (Herrmann, 2020). Implementing the interaction between maintenance actors and the predictive maintenance system is a critical challenge in achieving this, which has not been sufficiently addressed. While interaction models have been developed that indicate how maintenance actors should be involved (Herrmann, 2020), there remains a research gap regarding interaction modes and the appropriate supporting technology.

3. RELATED WORK

This section looks at related work regarding the state-of-the-art in human intervention in predictive maintenance and DIAs.

3.1 Human Intervention in Predictive Maintenance Systems

Interaction with predictive maintenance systems often occurs via computer screens or increasingly mobile devices (Börütecene and Löwgren, 2020). Visualisation of diagnosis data, prediction curves, and maintenance recommendations is often complex and difficult to quickly understand, especially for staff not experienced in digital technologies (Roda, Macchi and Fumagalli, 2018). While AI can support and personalise visualisation, which can help focus complex data views on personal needs and maintain situational awareness (Börütecene and Löwgren, 2020), there remain problems with screen-based interfaces. Needing to access a computer to perform maintenance tasks is obtrusive and inefficient. Operators are often forced to put away mobile devices where they cannot be damaged when they work (Havard *et al.*, 2016).

There is consequently a need for **advanced interaction modes** with predictive maintenance systems. AR is one, which is gaining traction today. Operators can use it to see maintenance information superimposed upon the asset-under-maintenance in an unobtrusive and potentially hands-free way. However, a case study shows that maintenance tasks may be performed faster with explanations on paper than through an AR system (Syberfeldt *et al.*, 2015). Also, operators may not correctly understand the augmentations displayed (Havard *et al.*, 2016). The same study also shows that humans better understand maintenance procedures if multiple channels present visual and audio information. There remains a need for an appropriate or at least supplementary interaction mode for predictive maintenance users, which does not interfere with maintenance tasks but fulfils the users' needs for rich, intuitive, and unobtrusive information and interaction.

3.2 Digital Intelligent Assistants

An assistant supports or entirely takes over time-consuming, stressful, or otherwise not desirable activities for the client (Wellsandt, Foosherian and Thoben, 2020). A digital assistant (DA) is software that achieves the same goals for one or more users. Besides, it is a type of **conversational agent** interacting with one or more users via natural language (Harms et al., 2019). Conversational agents include voice-user-interfaces (VUIs) agents, text-based agents (Chatbots), and embodied conversational agents. This paper focuses on the VUI type agent. We refer to it as **voice-enabled DIA** to stress that it is more than a human-computer interface. Emphasizing its "intelligence" is necessary because most of an assistant's utility benefits from its knowledge about, for instance, the surrounding, supported tasks, and background of its users.

A DIA possesses skills that the user perceives as difficult, such as calculating or gathering information in a short time. From a computer science viewpoint, these skills require little actual intelligence. Typically, they depend on rule execution, which is a domain where software generally outperforms humans. Users, however, typically associate these skills with intelligence.

The technological research behind voice-enabled DIA's reaches back to the first interactive voice response (IVR) systems in the 1970s – some principles of dialog design are still valid since then (Pearl, 2016). A voice-enabled DIA has five essential **components** (Wellsandt, Foosherian and Thoben, 2020). The *speech-to-text* (STT) component transcribes a user's audio utterances. *Natural language understanding* (NLU) extracts semantic elements from this transcribed text (Ram et al., 2018), typically through data analytics processes built with machine learning. Essential semantic elements are the user's intents and entities. Intents are what the user wants the assistant to do. Entities are pieces of information that the assistant needs to know to respond meaningfully (Coates, 2019). Examples for typical entities include persons, locations, items, dates, and activities. The *dialog management* (DM) component analyses the extracted intents and entities and decides how the assistant should respond. During this analysis, the assistant often has to integrate information stored in *external software*. DM functions can request information from external software that is too difficult to create by the assistant; it can also trigger the DM proactively. Access to external software is critical for the assistant's capabilities and perceived intelligence. Finally, the *text-to-speech* (TTS) component turns the generated text into a computer-generated voice.

4. HUMAN INTERVENTION IN PREDICTIVE MAINTENANCE PROCESSES

Fig. 1 shows a five-phase, unified model of predictive maintenance synthesizing concepts of predictive maintenance, proactive computing (Etzion, 2016), Gartner's levels of industrial analytics maturity (Eriksen and Steenstrup, 2015) and ISO 13374/MIMOSA OSA-CBM (MIMOSA, 2008). The next sections contain a brief description of each phase followed by an attempt at identifying potentials for human intervention

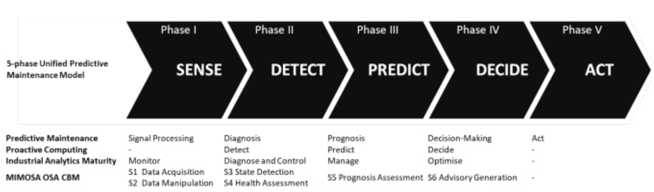

Fig. 1. Five-phase unified predictive maintenance model

The **sense phase** covers data generation, data acquisition, communication, and data pre-processing. Relevant data describes an asset's current condition and stems from sensors integrated into an asset. It can be accessed directly via an IoT hub or through systems such as SCADA or a Manufacturing Execution System (MES). By digitalising manufacturing assets, the physical world is connected to the virtual world (Lundgren, Bokrantz and Skoogh, 2020), which is the basis for predictive maintenance. The sense phase's output is typically a sensor data stream that is forwarded to the detect phase for further analysis. This data stream may be pre-processed on the network edge, for example, to clean the data.

Potential human interventions in the sense phase include *changing, configuring, controlling, and validating sensors* monitoring the assets (Truong, 2018). If abnormal measurements are shown, sensors need to be inspected to verify if they are working correctly or need to be replaced. Also, the system will need to be able to handle new sensors or other data sources made available after deployment (Herrmann, 2020). Edge computing requires *configuring, modifying, and verifying pre-processing algorithms* at the network edge.

The **detect phase** identifies abnormal behaviour by analysing incoming data streams. Different algorithms are applied here to detect the asset's current state. Detection and diagnosis are often used synonymously since failure causes may be derived from symptoms detected in the data stream. Abnormalities and patterns in the data are identified, failure modes then classify the patterns. The most basic classification in maintenance differentiates "faults" and "in order". Defect classification carries out pattern recognition by applying feature extraction methodology (Clarke, Duda and Hart, 1974). It often employs AI methods such as artificial neural networks (Kreisel, 2007). Patterns that have been classified require interpretation to understand the symptoms they signify. These interpretations need to be encoded into the predictive maintenance system for decisions to be taken based on the detection and diagnosis functionality.

Interpretation of classified patterns requires the expert knowledge of very experienced staff who understand the symptoms and identify the root causes (Moczulski, 2004). The potential for automating this process is limited (Cholewa, 2004), and further analysis might be required (Truong, 2018). This situation means that the intervention of experts is often required. Reliable diagnosis results are significant since uncertainties introduced here affect predicting future asset conditions, lead to false alarms, and ultimately influence the decision-making process (Hess *et al.*, 2006; Jardine, Lin and Banjevic, 2006; Patrick *et al.*, 2009). This means that *diagnosis results need to be verified*, which is challenging to automate (Vogl, Weiss and Helu, 2019). Regular expert interaction is required to *ensure the plausibility of detection and diagnosis results* (Herrmann, 2020).

The **predict phase** forecasts the condition and degradation of assets (prediction or prognosis), which is done by continuously observing sensor data streams and considering historical data (Denkena, Blümel and Kröning, 2010). It encompasses time-series analysis and prognosis and the estimation of Remaining Useful Life (RUL). RUL assumes that a physical system's quality decreases over time (Bousdekis *et al.*, 2019), and should include an estimated confidence interval. Methods of time-related and frequency analyses are used to analyse industrial machine data (Patrick *et al.*, 2009; Denkena, Blümel and Kröning, 2010; Vogl, Weiss and Helu, 2019). Multiple methods can be sequentially applied as cascade calculations to produce appropriate predictions. This phase's result is forwarded as an input to the decide phase, e.g., as a RUL calculation, degradation curve, or Weibull distribution.

The complexity of calculation flows in this phase can lead to intransparency regarding predictions generated by the system.

Results may also seem ambivalent and difficult to translate into a relevant business indicator such as RUL: like the decide phase, an expert must *interpret and verify* prognoses. The plausibility and trustworthiness of results are also critical to system acceptance by the users. Operators should be enabled to *explore the system and research it with what-if-questions or explore changing parameters, thresholds, or analysis methods* to overcome these deficits in the explainability of prediction results (Herrmann, 2020).

The **decide phase** deals with generating actionable decisions based on the output of the previous phases. Decision-making can be carried out by maintenance experts, who define maintenance actions based on their knowledge and experience. However, this requires maintenance experts to be proficient in interpreting the prediction algorithms' outputs, which means they need to be well-versed in engineering and data science. A preferable approach is for the predictive maintenance system to proactively recommend optimal actions and schedules that avoid or mitigate predicted failures (PricewaterhouseCoopers BWC, 2018; Bousdekis et al., 2019). However, the uncertainties involved in predictive analytics algorithms and degradation processes, combined with time constraints for decision-making, make the implementation of appropriate algorithm-driven approaches challenging. Solutions are often sector or use-case specific and only valid under certain assumptions.

Knowledge about appropriate maintenance needs to be introduced into the system as machine-processable rules. This assumes intense cooperation between maintenance experts and predictive maintenance system developers. However, even experienced engineers have little experience with these systems, making the system's deployment and configuration in specific use cases cost and time-intensive (Bousdekis et al., 2018). *Maintenance rules may need to be revised* if practices change or experience or knowledge shows more effective actions may be taken (Truong, 2018).

The **act phase** finally includes all actions taken based on the previous phase's decisions, for example, inspection, maintenance, verification, and reporting.

Maintenance operators require *information about how to carry out maintenance tasks*, for example, instructions about how to service a machine or replace and test a component. *Feedback about the success or failure of recommended maintenance actions* can help improve future recommendations.

5. SOLUTION CONCEPT

(Wellsandt et al., 2020) proposed a DIA to provide a fast, intuitive, and potentially hands-free interaction with predictive maintenance systems. It has basic, utility, and maintenance functions. This section uses the identified potentials described above and results from previous work to propose how DIAs could improve working with predictive maintenance systems. Fig. 2 summarizes the proposed improvement areas.

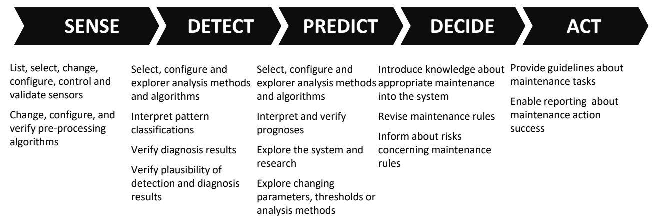

Fig. 2. Potential DIA support by predictive maintenance phase

Typical interventions in the **sense phase** are selecting and modifying data sources and the configuration of queries. The DIA user could request an overview of the available data sources and select relevant ones for analysis. If needed, the user could configure database queries by instructing the assistant to consider a specific timeframe or to filter a time series by defined business objects (e.g., assets, processes, or locations). The assistant would typically offer a form-structure that the user fills during a dialog.

In the **detect and predict phases**, the DIA support focuses on the selection, configuration, and question-and-answer-based exploration of analysis methods and algorithms. This support is beneficial when previously selected data sources do not meet specific methods' requirements (e.g., noisy data). The assistant can warn or prohibit using specific methods or algorithms. Furthermore, users may configure process parameters, such as thresholds via voice commands that are typically faster than typing – this helps to explore data. Finally, the user can verify the results of the analysis processes by confirming or denying that they have been observed on the physical asset.

The main support in the **decide phase** is that the DIA could allow maintenance coordinators to select, revise, add or delete recommendations given for specific detection and prediction results. Guidance may include warnings raised once a selection would increase, for instance, safety, security, or work-plan-related risks.

Finally, in the **act phase**, the assistant could offer support to maintenance technicians and machine operators in the field. This includes guidance how to perform inspections or repair, and access to context information during failure diagnosis (e.g. machine documentation and component states). This support can be part of an AR application for maintenance. Finally, the assistant could accept user feedback about the effectiveness of recommended maintenance activities. This feedback can contribute to supervised learning mechanisms that increase the accuracy of detection and prediction results.

Fig. 3 shows a simplified view of a generic, data-driven predictive maintenance system. The green lines show interactions between the DIA and the relevant system components.

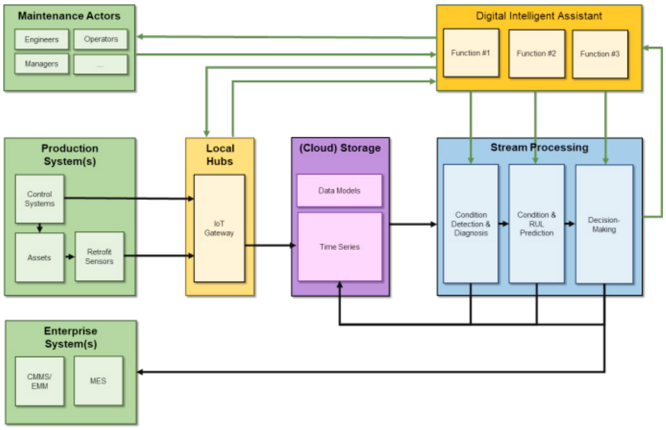

Fig. 3. Predictive Maintenance System Architecture with DIA

The human maintenance actors are involved in production and maintenance management, planning, optimisation and execution tasks. The **production system(s)** contains the assets under maintenance, comprising control systems and sensors, which provide real-time streaming data to the predictive maintenance system. The **enterprise systems** contribute to the management and execution of production and maintenance plans, which the decision-making results may modify. Relevant systems include Computerised Maintenance Management Systems (CMMS), Electronic Maintenance Modules (EMM), and MES. Streaming sensor data is collected via **local hubs** implementing IoT gateways to access and manage data generated by the production system's sensors. Streaming data is pushed into the predictive maintenance system, e.g., via a streaming message broker. The **(cloud) storage** element stores all types of data processed or generated by the predictive maintenance system. Both streaming sensor data and the intermediate results of diagnosis and prognosis analysis are stored here in a time series database. The **stream processing** component includes functionalities responsible for analysing stream data typically generated by sensors installed in the end user's assets and consists of a component for **condition detection & diagnosis** and one for **condition and RUL prediction**, providing forecasts for feature values of sensor data. **Decision-making** receives streams of predictions about future failure modes or RUL estimations and generates optimal maintenance prescriptions based on that and other maintenance planning factors. All components exchange data via a Kafka message bus. The DIA connects with the overall system in the same way.

6. CONCLUSIONS

DIAs will take a unique place in socio-technical predictive maintenance systems because they can offer intuitive and unobtrusive assistance in all predictive maintenance phases. Companies can use DIAs with different hardware, such as smart speakers, tablets, smartphones, desktop computers, and mixed reality applications. This flexibility allows digital assistants to address workers' requirements in maintenance tasks, such as hands-free interaction and fast access to information. Finally, deeper integration of humans into predictive maintenance systems creates opportunities for so-called hybrid-intelligence systems. In such future systems, humans and computers complement each other and evolve together. The next step in our research is to implement a fully functioning demonstrator and to evaluate it within an industrial use case. This demonstrator must also address general AI application challenges, such as an assistant's accountability, biased training data, and biased dialog structures.

ACKNOWLEDGEMENTS

This work is partly funded by the European Union's Horizon 2020 research and innovation programme via the projects UPTIME "Unified Predictive Maintenance System" (Grant agreement No 768634) and COALA "COgnitive Assisted agile manufacturing for a LAbor force supported by trustworthy Artificial Intelligence" (Grant agreement No 957296) as well as the EIT Manufacturing projects RAMEN "Resilient Industrial Platform for the Advanced Visualisation of Predictive Maintenance" and DIAMOND "Digital Intelligent Assistant for Predictive Maintenance as a Response to Demanding Employee Skill Requirements".

REFERENCES

Börütecene, A. and Löwgren, J. (2020) 'Designing human-automation collaboration for predictive maintenance', in DIS 2020 Companion - Companion Publication of the 2020 ACM Designing Interactive Systems Conference. New York, NY, USA: Association for Computing Machinery, Inc, pp. 251–256. doi: 10.1145/3393914.3395863.

Bousdekis, A. et al. (2015) 'A proactive decision making framework for condition-based maintenance', Industrial Management and Data Systems, 115(7), pp. 1225–1250. doi: 10.1108/IMDS-03-2015-0071.

Bousdekis, A. et al. (2018) 'Review, analysis and synthesis of prognostic-based decision support methods for condition based maintenance', Journal of Intelligent Manufacturing. Springer New York LLC, pp. 1303–1316. doi: 10.1007/s10845-015-1179-5.

Bousdekis, A. et al. (2019) 'Decision making in predictive maintenance: Literature review and research agenda for industry 4.0', IFAC-PapersOnLine, 52(13), pp. 607–612. doi: 10.1016/j.ifacol.2019.11.226.

Cholewa, W. (2004) 'Expert Systems in Technical Diagnostics', in Fault Diagnosis. Springer Berlin Heidelberg, pp. 591–631. doi: 10.1007/978-3-642-18615-8_15.

Clarke, M. R. B., Duda, R. O. and Hart, P. E. (1974) 'Pattern Classification and Scene Analysis.', Journal of the Royal Statistical Society. Series A (General). JSTOR, 137(3), p. 442. doi: 10.2307/2344977.

Coates, D. A. (2019) Voice Applications for Alexa and Google Assistant. Shelter Island, NY: Manning Publications Co. Available at: https://cds.cern.ch/record/2691774 (Accessed: 27 November 2020).

Denkena, B., Blümel, P. and Kröning, S. (2010) 'Simulationsbasierte Instandhaltung. Optimierung der Instandhaltungsplanung und -steuerung mit Hilfe ereignisorientierter Simulation.', wt Werkstattstechnik online, 100(7/8), pp. 598–603.

Harms, J. G. et al. (2019) 'Approaches for dialog management in conversational agents', IEEE Internet Computing. Institute of Electrical and Electronics Engineers Inc., 23(2), pp. 13–22. doi: 10.1109/MIC.2018.2881519.

Havard, V. et al. (2016) 'Augmented industrial maintenance (AIM): A case study for evaluating and comparing with paper and video media supports', in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). Springer Verlag, pp. 302–320. doi: 10.1007/978-3-319-40621-3_22.

Herrmann, T. (2020) 'Socio-technical design of hybrid intelligence systems – The case of predictive maintenance', in Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). Springer, pp. 298–309. doi: 10.1007/978-3-030-50334-5_20.

Hess, A. et al. (2006) 'Challenges, issues, and lessons learned chasing the "big P": Real predictive prognostics part 2', in IEEE Aerospace Conference Proceedings. IEEE, pp. 1–19. doi: 10.1109/AERO.2006.1656124.

Jardine, A. K. S., Lin, D. and Banjevic, D. (2006) 'A review on machinery diagnostics and prognostics implementing condition-based maintenance', Mechanical Systems and Signal Processing. Academic Press, pp. 1483–1510. doi: 10.1016/j.ymssp.2005.09.012.

Kamar, E. (2016) 'Directions in hybrid intelligence: Complementing AI systems with human intelligence', in IJCAI International Joint Conference on Artificial Intelligence, pp. 4070–4073. Available at: https://www.microsoft.com/en-us/research/wp-content/uploads/2016/11/hi.pdf (Accessed: 30 November 2020).

Kreisel, D. (2007) A Brief Introduction to Neural Networks, dkriesel.com. Available at: http://www.dkriesel.com/_media/science/neuronalenetze-en-zeta2-2col-dkrieselcom.pdf (Accessed: 11 September 2020).

Lundgren, C., Bokrantz, J. and Skoogh, A. (2020) 'Performance indicators for measuring the effects of Smart Maintenance', International Journal of Productivity and Performance Management. Emerald Group Publishing Ltd. doi: 10.1108/IJPPM-03-2019-0129.

McCarthy, J., Zinser, M. and Spindelndreier, D. (2013) The Maintenance Advantage in Manufacturing, Boston Consulting Group Webpage. Available at: https://www.bcg.com/publications/2013/lean-maintenance-advantage-manufacturing-achieving-excellence-three-dimensions (Accessed: 8 September 2020).

Moczulski, W. (2004) 'Methods of Acquisition of Diagnostic Knowledge', in Fault Diagnosis. Springer Berlin Heidelberg, pp. 675–718. doi: 10.1007/978-3-642-18615-8_17.

Patrick, R. et al. (2009) 'Diagnostic enhancements for air vehicle HUMS to increase prognostic system effectiveness', in IEEE Aerospace Conference Proceedings. doi: 10.1109/AERO.2009.4839653.

Pearl, C. (2016) Designing Voice User Interfaces: Principles of Conversational Experiences - Cathy Pearl - Google Books, New Media and Society. O'Reilly Media, Inc.

Peng, Y., Dong, M. and Zuo, M. J. (no date) 'Current status of machine prognostics in condition-based maintenance: a review', Springer. doi: 10.1007/s00170-009-2482-0.

Ram, A. et al. (2018) 'Conversational AI: The Science Behind the Alexa Prize', arXiv. arXiv. Available at: http://arxiv.org/abs/1801.03604 (Accessed: 27 November 2020).

Roda, I., Macchi, M. and Fumagalli, L. (2018) 'The future of maintenance within industry 4.0: An empirical research in manufacturing', in IFIP Advances in Information and Communication Technology. Springer New York LLC, pp. 39–46. doi: 10.1007/978-3-319-99707-0_6.

Syberfeldt, A. et al. (2015) 'Visual Assembling Guidance Using Augmented Reality', in Procedia Manufacturing. Elsevier B.V., pp. 98–109. doi: 10.1016/j.promfg.2015.09.068.

Truong, H. L. (2018) 'Integrated Analytics for IIoT Predictive Maintenance Using IoT Big Data Cloud Systems', in Proceedings - 2018 IEEE International Conference on Industrial Internet, ICII 2018. Institute of Electrical and Electronics Engineers Inc., pp. 109–118. doi: 10.1109/ICII.2018.00020.

Vogl, G. W., Weiss, B. A. and Helu, M. (2019) 'A review of diagnostic and prognostic capabilities and best practices for manufacturing', Journal of Intelligent Manufacturing. Springer New York LLC, 30(1), pp. 79–95. doi: 10.1007/s10845-016-1228-8.

Wellsandt, S. et al. (2020) 'Concept of a Voice-Enabled Digital Assistant for Predictive Maintenance in Manufacturing', in TESConf 2020 - 9th International Conference on Through-life Engineering Services. Cranfield, UK: Elsevier BV, p. 8. doi: 10.2139/ssrn.3718008.

Wellsandt, S., Foosherian, M. and Thoben, K.-D. (2020) 'Interacting with a Digital Twin using Amazon Alexa', Procedia Manufacturing.